

Predicting Future Bike Accident Hotspots in Amsterdam

Using HDBSCAN Clustering & Multi-Source Data Analysis

Esra Demirel
31.05.25

Data-Driven Approach



Methodology

Introduction to Bike Culture in Amsterdam



PREDICTION MODEL



Mapping Accident Hotspots



Key Findings Future Enhancements



Introduction to Bike Culture in Amsterdam

Cycling in the Netherlands: More Than Transport

Cycling is a key part of Dutch culture.

- About 27% of trips are by bike.
- Healthy lifestyle and environment friendly.
- Amsterdam has over 500 km of bike paths.
- City design focuses on safety and easy access.
- Great cycling infrastructure.



Urban Cycling Safety: Ongoing Challenges

- **Intersections** remain key accident hotspots
- Rising cycling traffic increases potential risks
- Need for enhanced safety measures & smarter infrastructure
- Focus on data-driven urban planning for future improvements



Cycling in the Netherlands: More Than Transport

Cycling is a key part of Dutch culture.

- About 27% of trips are by bike.
- Healthy lifestyle and environment friendly.
- Amsterdam has over 500 km of bike paths.
- City design focuses on safety and easy access.
- Great cycling infrastructure.



Urban Cycling Safety: Ongoing Challenges

- **Intersections** remain key accident hotspots
- Rising cycling traffic increases potential risks
- Need for enhanced safety measures & smarter infrastructure
- Focus on data-driven urban planning for future improvements



Predicting Future Bike Accident Hotspots in Amsterdam

Using HDBSCAN Clustering & Multi-Source Data Analysis

Esra Demirel
31.05.25

Data-Driven Approach



Methodology

Introduction to Bike Culture in Amsterdam



PREDICTION MODEL



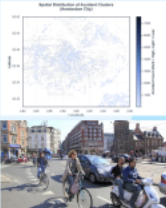
Mapping Accident Hotspots



Key Findings Future Enhancements



Data-Driven Approach



Introduction to Open Datasets

29,763 bike accidents analyzed (2014-2023)

- Goal: Predict future accident hotspots
- Method: HDBSCAN clustering + weather + injury data
- Output: 2025 risk predictions for Amsterdam

Key Statistics

- 29,763 total accidents
- 170 hotspot clusters identified
- 1,461 injury accidents
- 34 fatal accidents
- 95% accident increase (2014-2023, 10years)

Impact: Every prediction saves lives



Data Sources and Collection Methods

Three Independent Datasets:

- Coordinates:** 29,763 accident records with GPS
- Weather:** 7 years temperature, precipitation, wind
- Injury Data:** Severity + vulnerability patterns (1,461 incidents)

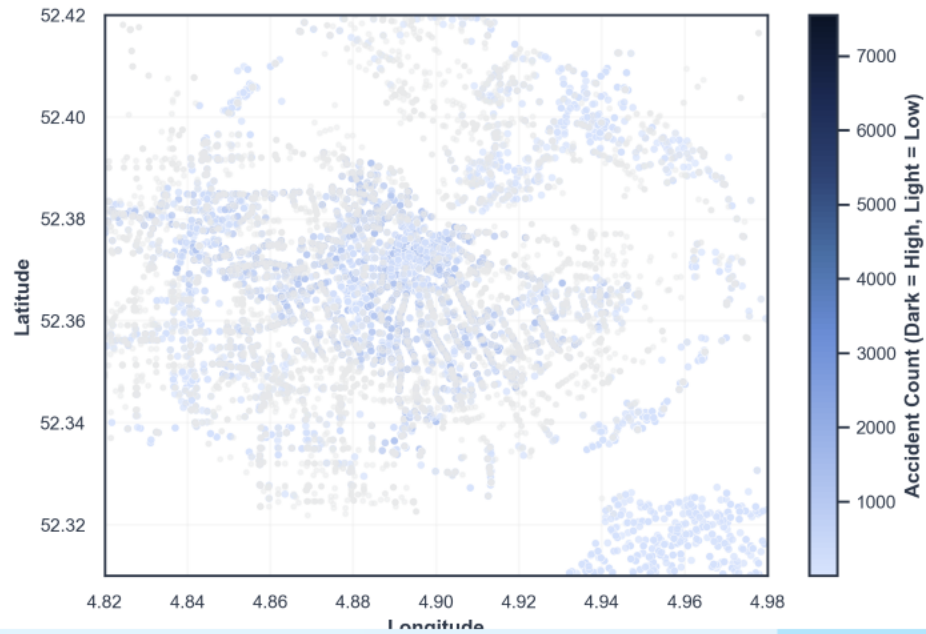
Key: No data merging - preserved data integrity

METHODOLOGY

- 01** Clean coordinates - Amsterdam area filter
- 02** Cluster with HDBSCAN - 170 hotspots found
- 03** Integrate weather risk factors
- 04** Analyze injury severity patterns
- 05** Predict 2025 hotspot risks

Quality Silhouette Score 0.888

Spatial Distribution of Accident Clusters
(Amsterdam City)



Introduction to Open Datasets

29,763 bike accidents analyzed (2014-2023)

- **Goal:** Predict future accident hotspots
- **Method:** HDBSCAN clustering + weather + injury data
- **Output:** 2025 risk predictions for Amsterdam

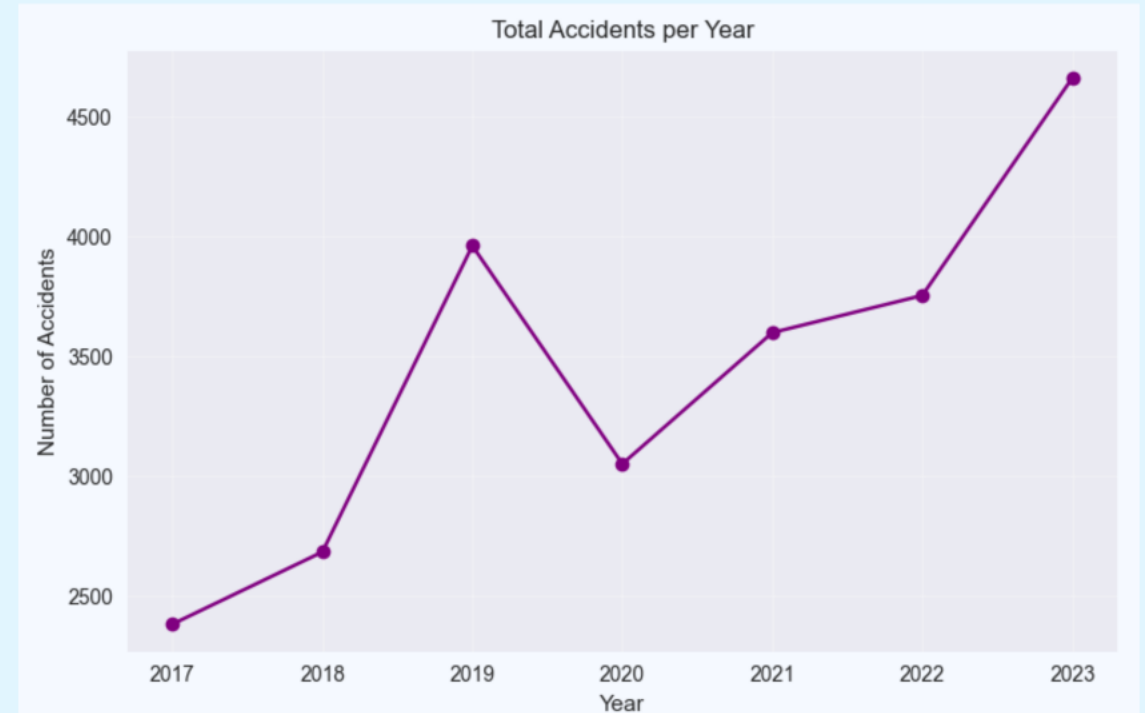


Key Statistics



- **29,763** total accidents
- **170** hotspot clusters identified
- **1,461** injury accidents
- **34** fatal accidents
- **95%** accident increase (2014-2023, 10years)

Impact: Every prediction saves lives





Data Sources and Collection Methods

Three Independent Datasets:



Coordinates: 29,765 accident records with GPS



Weather: 7 years temperature, precipitation, wind



Injury Data: Severity + weekday patterns (1,803 locations)

Key: No data merging - preserved data integrity

METHODOLOGY

01

Clean coordinates Amsterdam area filter

02

Cluster with HDBSCAN 170 hotspots found

03

Integrate weather risk factors

04

Analyze injury severity patterns

05

Predict 2025 hotspot risks

Quality: Silhouette Score 0.486

Predicting Future Bike Accident Hotspots in Amsterdam

Using HDBSCAN Clustering & Multi-Source Data Analysis

Esra Demirel
31.05.25



Mapping Accident Hotspots

CLUSTERING RESULTS

Performance Metrics:

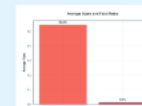
- 170 hotspot clusters identified
- 9,986 noise points (isolated incidents)
- Largest hotspot: 1,189 accidents (Ringweg-West)
- Average per cluster: 116.3 accidents



WEATHER IMPACT

Temperature: 10.6°C - 12.0°C
Precipitation: 1.5mm - 3.0mm average
Wind: 4.8m/s - 5.2m/s
Visibility: 22,500m - 23,400m

Accident Growth:
2017: 2,384 accidents
2023: 4,160 accidents
Growth 95% increase



INJURY SEVERITY

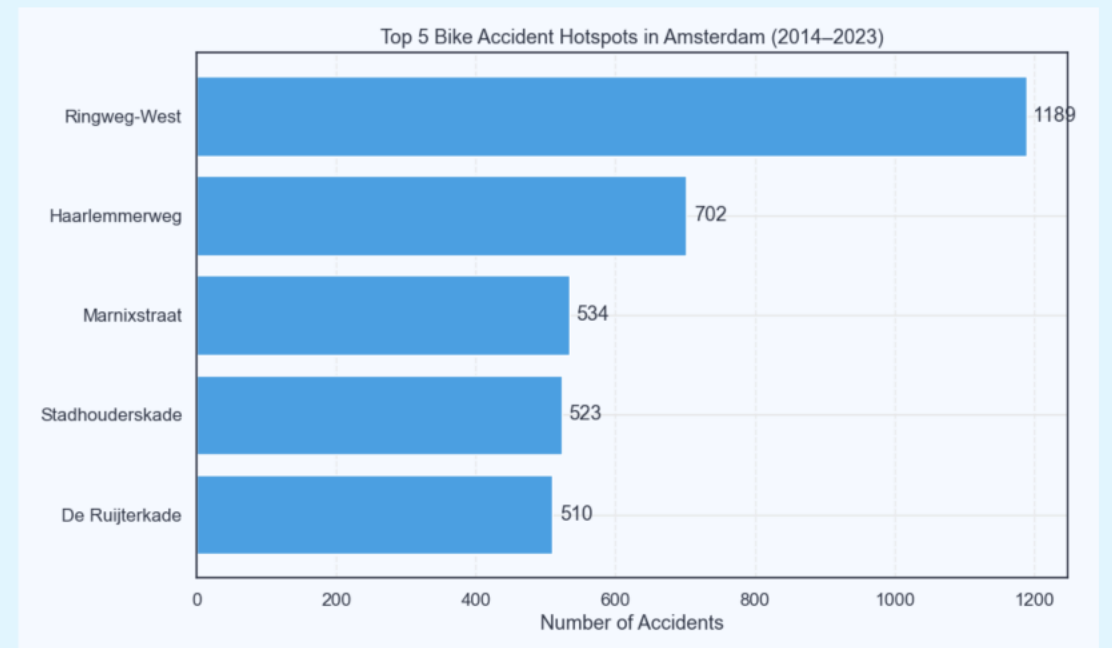
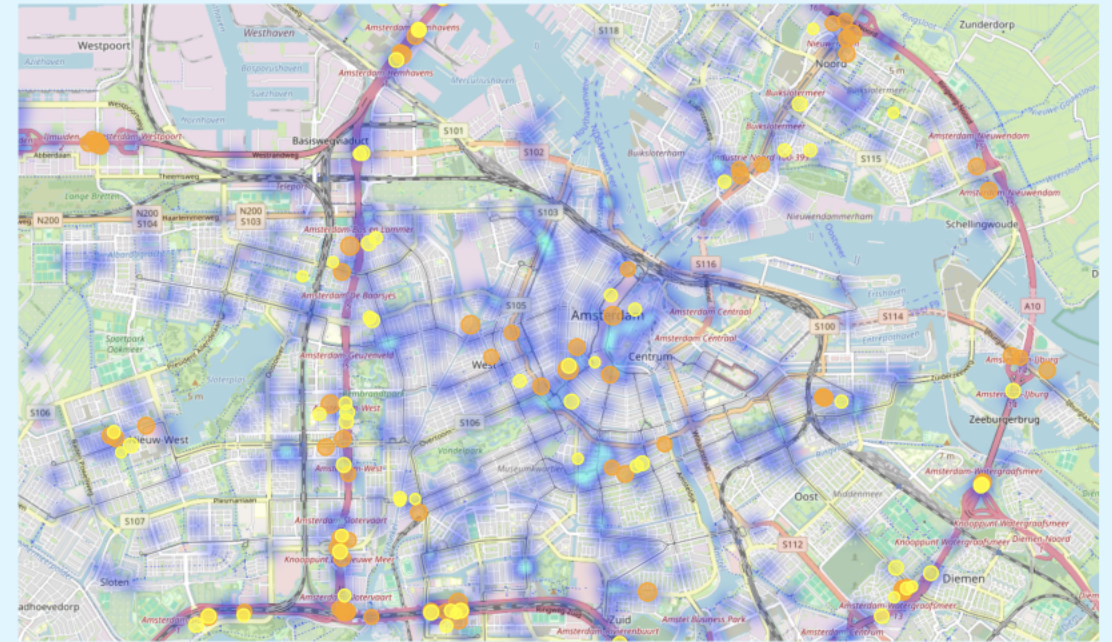
Severity Metrics:

- 56.4% average injury rate
- 1.57% total accident rate
- Fatal = 3 • Injury = 2 • Damage = 1

CLUSTERING RESULTS

Performance Metrics:

- 170 hotspot clusters identified
- 9,986 noise points (isolated incidents)
- Largest hotspot: 1,189 accidents (Ringweg-West)
- Average per cluster: 116.3 accidents

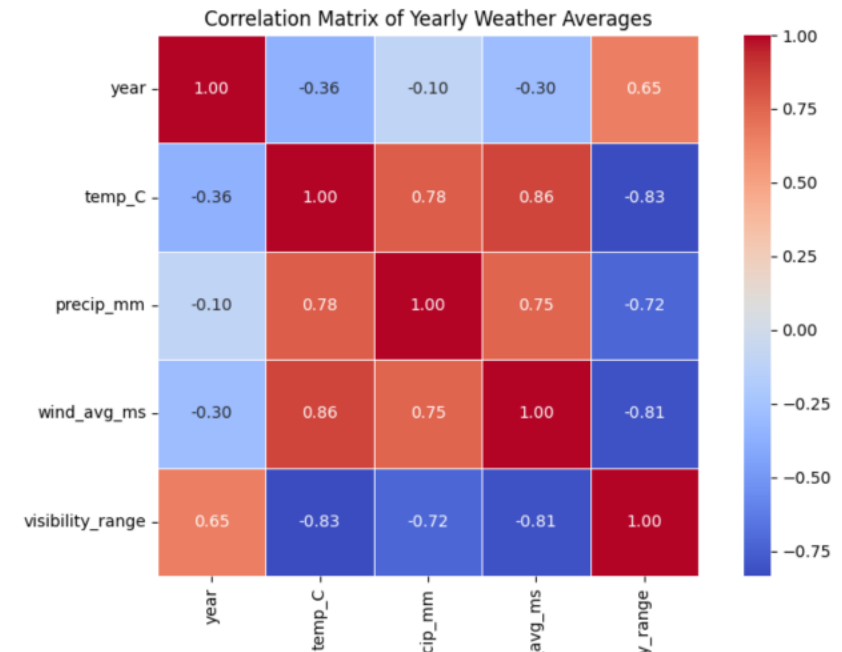


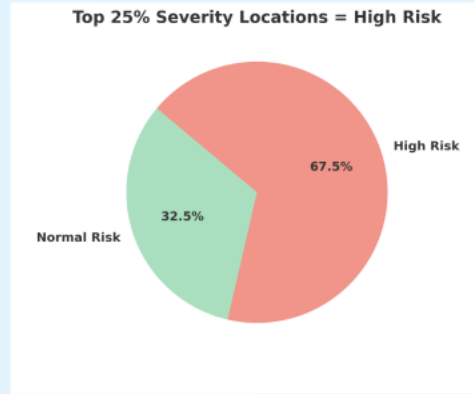
WEATHER IMPACT

Accident Growth:
2017: 2,384 accidents
2023: 4,660 accidents
Growth: 95% increase



Temperature: 10.6°C - 12.0°C
Precipitation: 1.5mm - 3.0mm average
Wind: 4.8m/s - 5.2m/s
Visibility: 22,500m - 23,400m

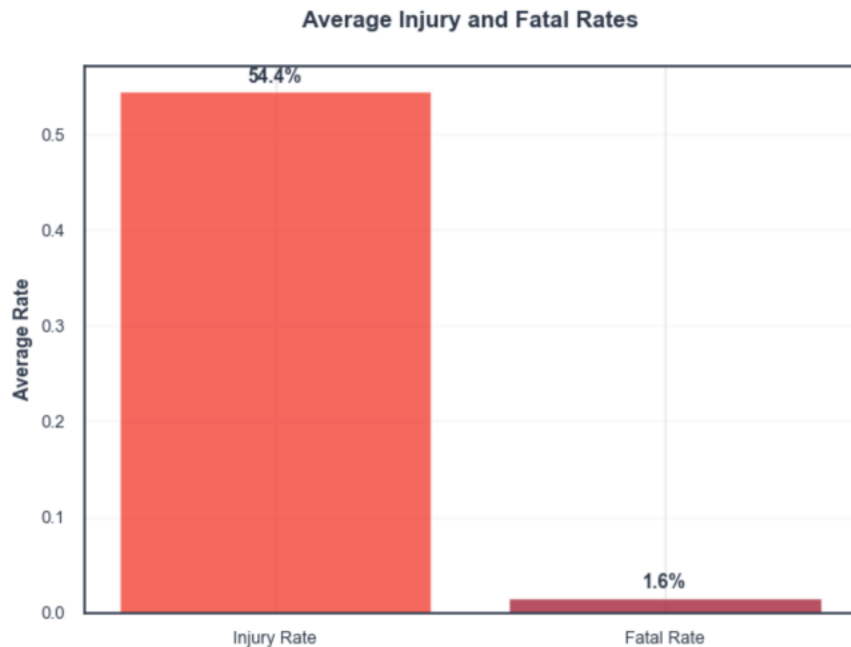




INJURY SEVERITY

Severity Metrics:

- 54.4% average injury rate
- 1.57% fatal accident rate
- $\text{Fatal} \times 3 + \text{Injury} \times 2 + \text{Damage} \times 1$



Predicting Future Bike Accident Hotspots in Amsterdam

Using HDBSCAN Clustering & Multi-Source Data Analysis

Esra Demirel
31.05.25

Data-Driven Approach



Methodology

Introduction to Bike Culture in Amsterdam



PREDICTION MODEL



Mapping Accident Hotspots



Key Findings Future Enhancements



PREDICTION MODEL

Scoring Formula (Weighted):

- Base Risk (40%): Historical accident frequency
- Weather Risk (30%): Temperature, precipitation, wind
- Temporal Weight (20%): Recent years emphasis
- Injury Risk (10%): High-severity proximity

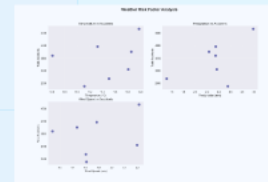


Risk Levels
High: Score > 0.7
Medium: Score 0.4-0.7
Low: Score < 0.4

HARSH WEATHER SCENARIO

Weather Impact:

- Temperature: -10% decrease
 - Precipitation: +30% increase
 - Wind: +20% increase
- Risk Changes:
- Average increase: +0.037 points
 - Top score: 0.950 (Cluster 117)
 - All hotspots: Elevated risk



2025 PREDICTIONS -NORMAL

Top Predicted Hotspots:

1. **Cluster 117** - Ringweg-West (0.555)
2. **Cluster 1** - Goosendamweg (0.532)
3. **Cluster 162** - Haarlemmerweg (0.514)

Results Summary:

- 0 High-risk predictions
- 94 Medium-risk predictions
- 76 Low-risk predictions



Scoring Formula (Weighted):

- Base Risk (40%): Historical accident frequency
- Weather Risk (30%): Temperature, precipitation, wind
- Temporal Weight (20%): Recent years emphasis
- Injury Risk (10%): High-severity proximity



Risk Levels:
High: Score > 0.7
Medium: Score 0.4-0.7
Low: Score < 0.4

2025 PREDICTIONS -NORMAL

Top Predicted Hotspots:

1. **Cluster 117** - Ringweg-West (0.555)
2. **Cluster 1** - Gaasperdammerweg (0.532)
3. **Cluster 162** - Haarlemmerweg (0.514)

Results Summary:

- **0** High-risk predictions
- **94** Medium-risk predictions
- **76** Low-risk predictions



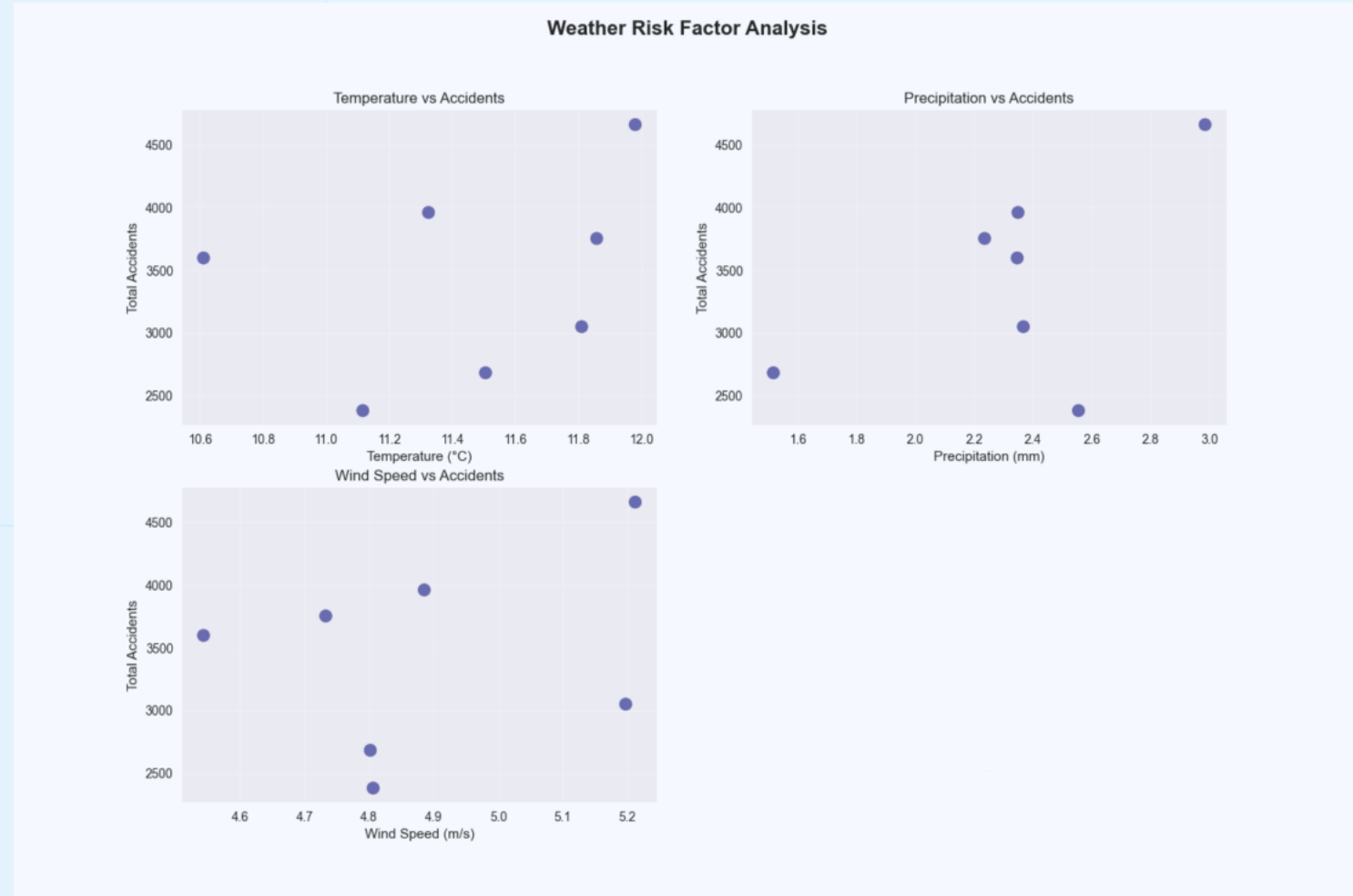
HARSH WEATHER SCENARIO

Weather Impact:

- Temperature: -10% decrease
- Precipitation: +30% increase
- Wind: +20% increase

Risk Changes:

- Average increase: +0.037 points
- Top score: 0.592 (Cluster 117)
- All hotspots: Elevated risk



Predicting Future Bike Accident Hotspots in Amsterdam

Using HDBSCAN Clustering & Multi-Source Data Analysis

Esra Demirel
31.05.25

Data-Driven Approach



Methodology

Introduction to Bike Culture in Amsterdam



PREDICTION MODEL



Mapping Accident Hotspots



Key Findings Future Enhancements



Key Findings Future Enhancements

Major Discoveries

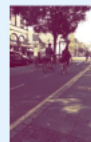
- 66% of accidents in 170 specific clusters
- 95% accident increase (2017-2023)
- 5.7% risk increase in harsh weather
- Ringweg-West: 261 total accidents



Key Achievement

Transforming 30,000 accidents into actionable safety intelligence

Thank You - Questions Welcome



Actionable Intelligence:

Immediate Action:
Monitor risk in real-time hotspots
Identify changing threat characteristics
Early warning to residents
Targeted safety campaigns

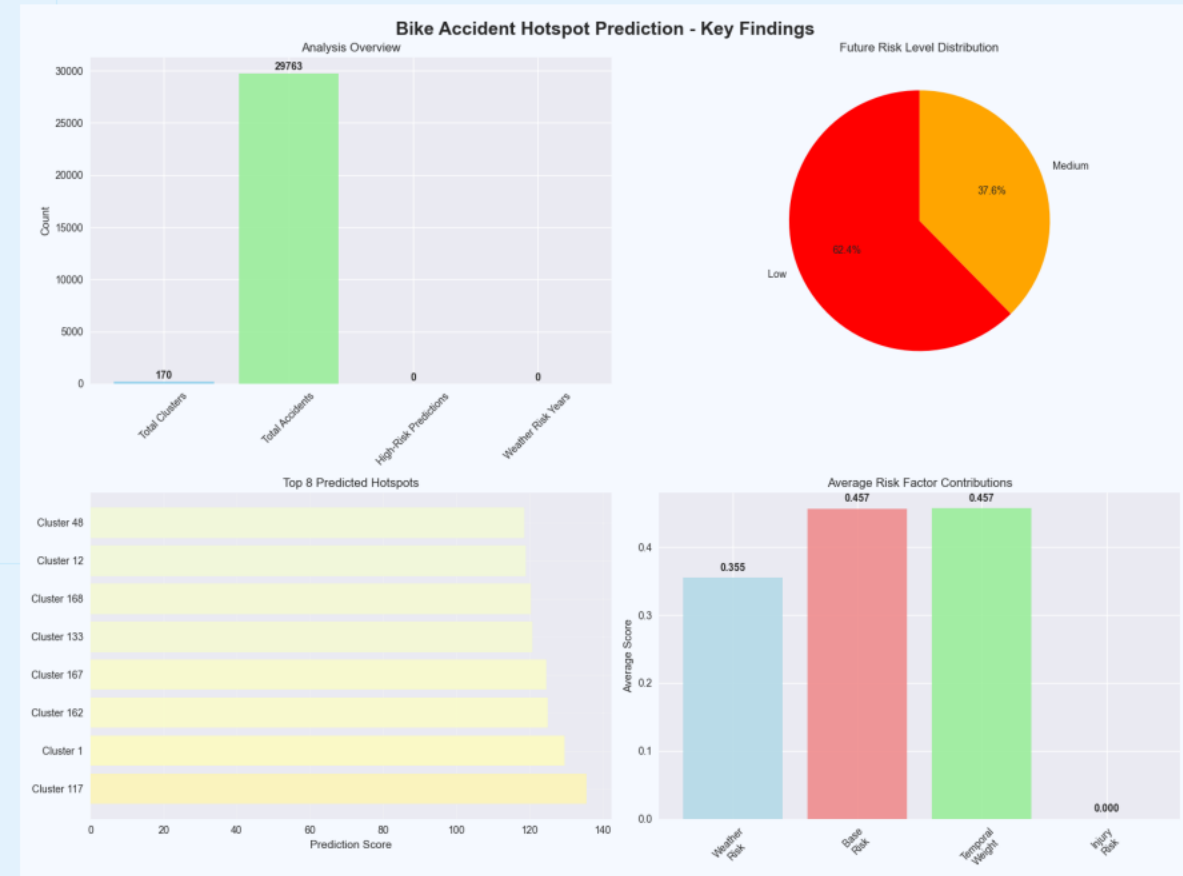
Future Enhancements:
Real-time traffic integration
Weather prediction models
Machine learning
Export to other Dutch cities

Research Extensions:
Machine learning models
Driver factor integration
Emergency response optimization



Major Discoveries

- 66% of accidents in 170 specific clusters
- 95% accident increase (2017-2023)
- 6.7% risk increase in harsh weather
- Ringweg-West: 261 total accidents





Actionable Intelligence:

Immediate Actions:

- Monitor top 10 predicted hotspots
- Enhance Ringweg-West infrastructure
- Deploy weather-based alerts
- Target safety campaigns

Future Enhancements:

- Real-time traffic integration
- Monthly prediction models
- Mobile cyclist app
- Expand to other Dutch cities

Research Extensions:

- Machine learning models
- Social factor integration
- Emergency response optimization

Key Achievement

Transforming 30,000 accidents into
actionable safety intelligence

Thank You - Questions Welcome



Predicting Future Bike Accident Hotspots in Amsterdam

Using HDBSCAN Clustering & Multi-Source Data Analysis

Esra Demirel
31.05.25

Data-Driven Approach



Methodology

Introduction to Bike Culture in Amsterdam



PREDICTION MODEL



Mapping Accident Hotspots



Key Findings Future Enhancements

